Washington State University

Daily Temperature Forecasting

Seasonal Regression with ARMA Errors

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Research objectives and hypotheses

Research question

Does combining annual seasonality with short-term residual dependence improve daily temperature forecasts?

- **H1:** The components provide complementary predictive information.
- **H2:** The model yields informative prediction intervals.

Forecast targets

$$\widehat{y}_{t+1|t} \qquad \widehat{y}_{t+7|t}$$

Daily mean temperature, °C

Evaluation
RMSE and MAE
Coverage and interval width

The empirical contribution is a controlled comparison across horizons.

Related work and study contribution

Dynamic harmonic regression Hyndman & Athanasopoulos, 2021

Fourier seasonality with ARMA errors for short-term dependence.

Systematic model selection Hyndman & Khandakar, 2008

A bounded ARIMA grid with a common chronological validation protocol.

Present study

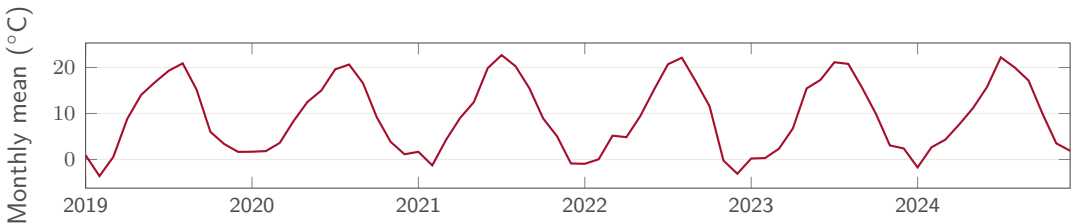
An audited NOAA station and a controlled comparison of seasonal structure and residual dynamics at two horizons.

References: otexts.com/fpp3/dynamic-regression.html; doi.org/10.18637/jss.v027.i03.

Data and chronological partitions

Pullman/Moscow Regional Airport
NOAA GSOD station 72785794129
2,191 observed days, 2019–2024
One missing day: December 31, 2021
Preserve the calendar. No interpolation.

Partition	Calendar	Observed
Train	1,461	1,460
Validation	365	365
Test	366	366



Dynamic harmonic regression

$$y_t = \underbrace{s_t}_{\text{annual seasonality}} + \underbrace{u_t}_{\text{correlated residual}}$$

$$s_t = \beta_0 + \sum_{k=1}^{K} \left[a_k \sin\left(\frac{2\pi k t}{365.25}\right) + b_k \cos\left(\frac{2\pi k t}{365.25}\right) \right]$$

$$\phi(B) u_t = \theta(B) \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2).$$

- Joint maximum-likelihood estimation in `statsmodels`.
- Known future calendar terms plus recursive residual forecasts.
- Selected specification: $K = 3$, ARMA$(1, 1)$ errors.

Rolling-origin evaluation

Estimation	Selection	Held-out test
2019–2022	2023 validation	2024
Fit each candidate on historical observations.	Minimize mean RMSE across the two horizons.	Refit through 2023, then freeze all coefficients.

$$\widehat{y}_{t+h|t} = \widehat{s}_{t+h} + \widehat{u}_{t+h|t}, \qquad h \in \{1, 7\}$$

Issue forecasts first. Reveal the next observation and update states afterward.

Identical origin–target pairs across models. No tuning on 2024.

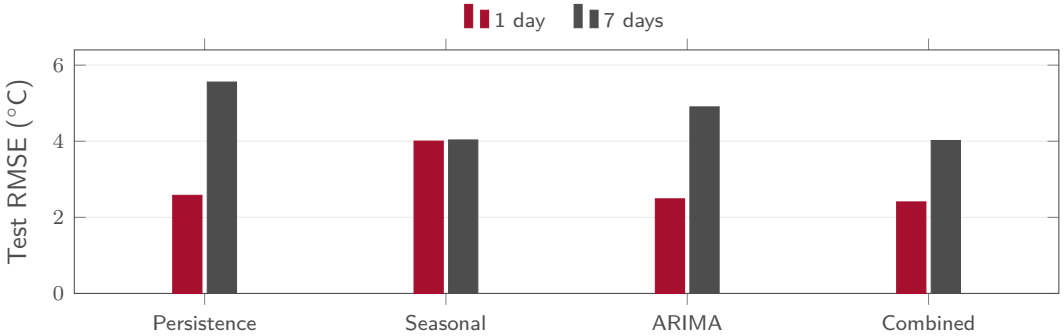
Model selection on 2023 validation

Model	RMSE (°C) 1 day	7 days
Persistence	2.441	5.248
Seasonal regression, $K = 3$	3.661	**3.672**
Nonseasonal ARIMA$(3, 0, 3)$	2.259	4.457
Combined: $K = 3$, ARMA$(1, 1)$	**2.169**	3.677

All 83 candidate fits converge.

The combined model leads at one day.
Seasonal regression is marginally better at seven days.

Held-out accuracy in 2024



Combined RMSE: 2.405°C at one day; 4.019°C at seven days.

Seasonal seven-day RMSE: 4.034°C. The combined gain is only 0.015°C.

Prediction intervals and calibration

Nominal coverage: 95%

Horizon	Coverage	Width (°C)
1 day	93.7%	9.295
7 days	95.6%	14.856

Residual diagnostics

Limited remaining mean dependence

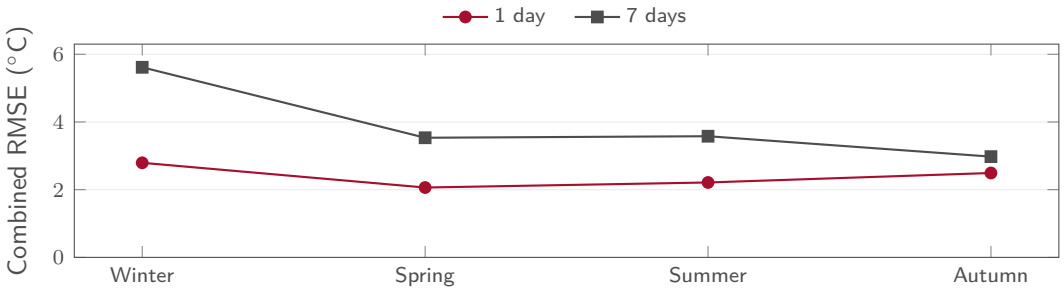
Nonnormal innovations

Evidence of changing variance

Point forecast accuracy does not establish interval calibration.

Gaussian intervals condition on fitted coefficients and omit parameter uncertainty.

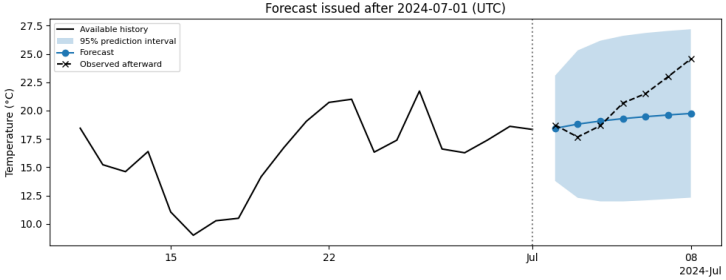
Seasonal performance and winter undercoverage



Seven-day winter coverage: 87.1%, compared with 95.6% annually.

Winter has the largest errors. Aggregate coverage conceals seasonal differences.

Reproducible Python demonstration



Notebook, section 6

Choose `ORIGIN`.

Filter available history with fixed parameters.

Check against the saved forecast and interval.

Demonstration origin: July 1, 2024. Observed future values appear only after forecasting.

Conclusions and future research

Main findings

- Residual dependence improves one-day predictions.
- Annual seasonality accounts for most seven-day performance.
- Winter undercoverage limits uncertainty interpretation.

Next steps

Additional stations and future test years; flexible variance models and innovation distributions; meteorological predictors and realistic data-release timing.

`https://github.com/lamanmirzayeva/Weather-Forecasting`